1 Long Range Minimal Models

1.1 General Formulation

The authors in [...] propose a method to study properties of long-ranged models in the vicinity of hypothetical cross-overs $s \lesssim s^\star$, at which (by hypothesis) the model should, in the infrared, transition from a long-ranged universality class to a short-ranged virasoro minimal models universality class $\mathcal{M}_{m+1,m}$ characterized by central charge and conformal weight given by Eq. (1.1). In other words, the authors hypothesize that special cross-over points $s^\star$ exist, at which the long-ranged model in the IR behaves as virasoro minimal models of type$\mathcal{M}_{m+1,m}$ and characterized by Eq. (1.1). By this hypothesis one should furthermore expect the emergence of multicritical points corresponding to different minimal models, as the parameter s gets tunned.

$$c = 1 - \frac{6}{m\left(m+1\right)} \; ; \qquad\qquad h_{r,s} = \frac{\left(\left(m+1\right)r - ms\right)^2 - 1}{4m\left(m+1\right)} \tag{1.1}$$

A dual-description in the form of Eq. (1.2) is proposed, where "..." indicates (potentially) additional self-interaction terms, and a short-ranged field ϕ_i corresponding to a primary of the virasoro moinimal model of type $\mathcal{M}_{m+1,m}$ is coupled to a long-ranged GFF χ by means of a weakly-relevant coupling parameter g. Given Δ_i the conformal weight of ϕ_i, s is tuned such that the conformal weight of the GFF is given by $\Delta_\chi = d - \Delta_i - \delta$ for a small parameter $0 < \delta << 1$. The operator $\mathcal{O} = \phi_i \chi$ will therefore have dimension $\Delta_\mathcal{O} = d - \delta$, so that the coupling g is weakly relevant, and χ is located at a region $s \lesssim s^\star$, that is, in the vicinity of the (hypothetical) cross-over to the short-range universality class characterized by $\mathcal{M}_{m+1,m}$.

$$S\left[\phi_i;\chi\right] = \int \frac{\left(\chi\left(x\right) - \chi\left(y\right)\right)^2}{\left|x-y\right|^{s+d}} dxdy + \int \partial_\mu \phi_i\left(x\right) \partial^\mu \phi_i\left(x\right) dx + g \int \chi\left(x\right) \phi_i\left(x\right) dx + \ldots \tag{1.2}$$

This description is motivated primarily by the works of [...]. The idea is that

1.2 Wetterich Equation

Starting from the action term specified by Eq. (1.2) above, one may propose an LPA' Ansatz for the running effective action specified by Eq. (1.3) below, where $U_k\left[\phi,\chi\right]$ is the potential term correpsonding to (potentially) additional self-interactions of ϕ and of χ.

$$\Gamma_k\left[\phi,\chi\right] = \frac{1}{2} Z_k^\phi p^2 \phi^2 + \frac{1}{2} Z_k^\chi p^s \chi^2 + U_k\left[\phi,\chi\right] \tag{1.3}$$

The problem is solved with an LPA' scheme, using Litim regulators. The solution is given by Eq. (1.4)The

Wetterich Equation can be expanded explicitly, Eq. (1.4).[1] The RG flow of the potential is correspondingly given by Eq. (1.5).[2]

$$\partial_k \Gamma_k [\phi, \chi] = \int \frac{(\partial_k R_{k,\phi}) \Gamma^R_{k,\chi\chi} + (\partial_k R_{k,\chi}) \Gamma^R_{k,\phi\phi}}{\Gamma^R_{k,\phi\phi}\Gamma^R_{k,\chi\chi} - g_k^2} \frac{d^2\vec{p}}{(2\pi)^2} \tag{1.4}$$

$$\partial_t \tilde{U}_t = -2\tilde{U}_t + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}_{t,\phi} + \frac{2 - s + \eta_\chi}{2}\tilde{\chi}\tilde{U}_{t,\chi} + \frac{C_\chi\left(1 + \tilde{U}_{t,\phi\phi}\right) + C_\phi\left(1 + \tilde{U}_{t,\chi\chi}\right)}{\left(1 + \tilde{U}_{t,\phi\phi}\right)\left(1 + \tilde{U}_{t,\chi\chi}\right) - g^2} \tag{1.5}$$

1.3 The $g \to 0$ Limit

In the limit of $g \to 0$, the denominator in Eq. (1.4) can be expanded in powers of g^2, Eq. (1.6).

$$\begin{aligned}\frac{C_\chi\left(1 + \tilde{U}_{t,\phi\phi}\right) + C_\phi\left(1 + \tilde{U}_{t,\chi\chi}\right)}{\left(1 + \tilde{U}_{t,\phi\phi}\right)\left(1 + \tilde{U}_{t,\chi\chi}\right) - g^2} &= \frac{C_\chi}{\left(1 + \tilde{U}_{t,\chi\chi}\right)} + \frac{C_\phi}{\left(1 + \tilde{U}_{t,\phi\phi}\right)} \\ &\quad - g^2 \frac{C_\chi\left(1 + \tilde{U}_{t,\phi\phi}\right) + C_\phi\left(1 + \tilde{U}_{t,\chi\chi}\right)}{\left(1 + \tilde{U}_{t,\phi\phi}\right)^2\left(1 + \tilde{U}_{t,\chi\chi}\right)^2} + \mathcal{O}\left(g^4\right)\end{aligned} \tag{1.6}$$

Namely one notices that in the $g \to 0$ limit, the equations (1) eventually decouple and the fields behave as independent long-range and short-range fields; and (2) even at $g \neq 0$, $g \to 0$, the perturbations can be linearized,

1.4 Mechanism of Crossover

The Wetterich Equation can be solved in a diagonalized basis. Namely, one considers the eigenproblem specified by Eq. (1.7) for the diagonalization of the running inverse propagator P_k^{-1}.

$$P_k^{-1}\vec{v} = \lambda_k \vec{v} \; ; \; P_k^{-1} = M D^{-1} M^{-1} \; ; \; D^{-1} = diag\begin{pmatrix} \frac{1}{\lambda_{k,+}} & \frac{1}{\lambda_{k,-}} \end{pmatrix} \; ; \; M = \begin{pmatrix} \vec{v}_+ & \vec{v}_- \end{pmatrix} \tag{1.7}$$

The Eigenvalues are obtained from solving the quadratic, Eq. (1.8).

$$\lambda^2 - \left(\Gamma^R_{k,\phi\phi} + \Gamma^R_{k,\chi\chi}\right)\lambda + \left(\Gamma^R_{k,\phi\phi}\Gamma^R_{k,\chi\chi} - g_k^2\right) = 0 \tag{1.8}$$

[1] *See* Appendix Appendix — Solution to the Wetterich Equation.
[2] *See* Appendix Appendix — Flow of the Potential.

The corresponding eigenvectors satisfy $\vec{v}_{\pm} = \begin{pmatrix} g_k \\ \lambda_{k,\pm} - \Gamma^R_{k,\phi\phi} \end{pmatrix}$, such that we build matrices, Eq. (1.9).

$$M = \begin{pmatrix} \vec{v}_+ & \vec{v}_- \end{pmatrix} = \begin{pmatrix} g_k & g_k \\ \lambda_{k,+} - \Gamma^R_{k,\phi\phi} & \lambda_{k,-} - \Gamma^R_{k,\phi\phi} \end{pmatrix} \quad ; \quad M^{-1} = \frac{1}{g_k\left(\lambda_{k,+} - \lambda_{k,-}\right)} \begin{pmatrix} \Gamma^R_{k,\phi\phi} - \lambda_{k,-} & g_k \\ \lambda_{k,+} - \Gamma^R_{k,\phi\phi} & -g_k \end{pmatrix} \tag{1.9}$$

Those matrices can be expressed in terms of a rotation matrix by means of the transformations Eq. (1.10).

$$tan\left(2\theta_M\right) = \frac{2g_k}{\Gamma^R_{k,\phi\phi} - \Gamma^R_{k,\chi\chi}} \; ; \qquad cos^2\left(\theta_M\right) = \frac{\Gamma^R_{k,\phi\phi} - \lambda_{k,-}}{\lambda_{k,+} - \lambda_{k,-}} \; ; \qquad sin^2\left(\theta_M\right) = \frac{\lambda_{k,+} - \Gamma^R_{k,\phi\phi}}{\lambda_{k,+} - \lambda_{k,-}} \; ; \tag{1.10}$$

The procedure leads to the proposed action, Eq. (1.2), be written in terms of two "synthetic", diagonalized oscillating nodes whose "inverse propagators" are given by the eigenvalues $\lambda_{\pm}$; and one furthermore identifies $\Gamma_{k,\psi_{\pm}} = \lambda_{k,\pm}$. Eq. (1.11).

$$S\left[\psi_+, \psi_-\right] = \int \left(\lambda_{k,+}\psi_+^2 + \lambda_{k,-}\psi_-^2\right) dp \tag{1.11}$$

The corresponding flows of the renormalization flux are given by Eqs. (1.12, 1.13).

$$\partial_k \lambda_{k,+} = \frac{k^{3-\eta_\phi}}{\lambda_{k,+}} \frac{1}{4\pi}\left(1 - \frac{\eta_\phi}{4}\right) cos^2\left(\theta_M\right) + \frac{k^{s+1} Z_k^\chi}{\lambda_{k,+}} \frac{s}{4\pi}\left(1 - \frac{\eta_\chi}{2+s}\right) sin^2\left(\theta_M\right) \tag{1.12}$$

$$\partial_k \lambda_{k,-} = \frac{k^{s+1} Z_k^\chi}{\lambda_{k,-}} \frac{s}{4\pi}\left(1 - \frac{\eta_\chi}{2+s}\right) cos^2\left(\theta_M\right) + \frac{k^{3-\eta_\phi}}{\lambda_{k,-}} \frac{1}{4\pi}\left(1 - \frac{\eta_\phi}{4}\right) sin^2\left(\theta_M\right) \tag{1.13}$$

One furthermore finds the flow of the mixing angle θ_M,Eq. (1.14).

$$\partial_k \theta_M = \frac{\left(\Gamma^R_{k,\phi\phi} - \Gamma^R_{k,\chi\chi}\right) \cdot \partial_k g_k - g_k \cdot \partial_k \left(\Gamma^R_{k,\phi\phi} - \Gamma^R_{k,\chi\chi}\right)}{\left(\Gamma^R_{k,\phi\phi} - \Gamma^R_{k,\chi\chi}\right)^2 - 4g_k^2} \tag{1.14}$$

1.5 Mechanism of the Crossover

One outlines here a few points concerning Eqs. (1.12, 1.13).

- First of all, one notices that Eqs. (1.12, 1.13) are exact, at least in the IR limit. Namely, no truncation assumptions were made with regards to the Ansatz in Eq. (1.3). One may raise very reasonable

concerns, namely because $\lambda_{k,\pm}$ is written as linear and non-linear combinations of $\Gamma^R_{k,\phi\phi}$ and $\Gamma^R_{k,\chi\chi}$; on the other hand, $\Gamma^R_{k,\phi\phi}$ and $\Gamma^R_{k,\chi\chi}$ have in principle different scaling dimensions; such that $\lambda_{k,\pm}$ (and by extension $\psi_\pm$) do not have well-defined scaling dimensions and therefore cannot (in general) represent a physical object. This furthermore implies that the "diagonalized" action object defined in Eq. (1.11), at least from a physical point of view, is generally "ill-defined" (more precisely, it is well-defined; but it does not generally represent a real physical object).

- This inconsistency can only be resolved, either by demanding that both dependancies attain the same dimensional scaling, or by demanding that one of the dimensional dependencies vanish. On the other hand, if those dependancies do not have the same dimensional scaling, it is precisely so, that one of the dependancies does in fact vanish in the IR limit; which is the regime that this equation is claimed to be valid for. Namely, one considers the IR limit $k \to 0$ and the scaling laws $\Gamma^R_{k,\phi\phi} \sim k^{2-\eta_\phi}$ and $\Gamma^R_{k,\chi\chi} \sim k^s$. One writes furthermore $s = s^\star \pm \delta$, where $s^\star = 2 - \eta_\phi$ is the Sak point. In this case, either $s > s^\star$, that is, $\delta > 0$, such that $\Gamma^R_{k,\chi\chi}$ falls-off faster than $\Gamma^R_{k,\phi\phi}$ and one identifies the LHS inequality in Eq. (1.15) below; or $s < s^\star$, that is, $\delta < 0$, such that $\Gamma^R_{k,\chi\chi}$ falls-off slower than $\Gamma^R_{k,\phi\phi}$ and one identifies the RHS inequality in Eq. (1.15). That is to say, in both cases $\lambda_{k,\pm}$ attains a well-defined physical dimension, which insofar as this extent is concerned resolves the purported inconsistency.

$$\lambda^{s>s^*}_{k,\pm} \sim \left(\frac{\Gamma^R_{k,\phi\phi} \pm \Gamma^R_{k,\phi\phi}}{2}\right) \pm \frac{g_k^2}{\Gamma^R_{k,\phi\phi}} + \mathcal{O}\left(g^4\right) \ ; \qquad \lambda^{s<s^*}_{k,\pm} \sim \left(\frac{\Gamma^R_{k,\chi\chi} \mp \Gamma^R_{k,\chi\chi}}{2}\right) \mp \frac{g_k^2}{\Gamma^R_{k,\chi\chi}} + \mathcal{O}\left(g^4\right) \tag{1.15}$$

- With this insight in hand one turns back to the flow equations, Eqs. (1.12, 1.13). Again one supposes $s = s^\star \pm \delta$ where $s^\star = 2 - \eta_\phi$ is the Sak point. In this case, within the flow of $\lambda_{k,+}$, the $cos^2(\theta_M)$ contribution falls as $k^{3-\eta_\phi}$ in the IR limit whereas the $sin^2(\theta_M)$ contribution falls as $k^{3-\eta_\phi \pm \delta}$ in the IR limit; such that when $s > s^\star$, that is, when $\delta > 0$, the sinusoidal contribution vanishes in the IR limit and only the cosinusoidal term surives; whereas on the other hand when $s < s^\star$, that is, when $\delta < 0$, the cosinusoidal contribution vanishes in the IR limit and only the sinusoidal contribution survives. Additionally, because one considers a weak coupling $g_k \to 0$, the sinusoidal term actually always vanishes whereas $cos^2(\theta_M) \to 1$. This analysis shows that $\partial_k \lambda_{k,+} \sim \frac{k^{3-\eta_\phi}}{\lambda_{k,+}} \frac{1}{4\pi}\left(1 - \frac{\eta_\phi}{4}\right) cos^2(\theta_M)$ when $s > s^\star$, and $\partial_k \lambda_{k,+} \sim 0$ when $s < s^\star$.

- The equivalent analysis carried out with respect to $\lambda_{k,-}$ yields the converse statement, that is to say, one obtains that $\partial_k \lambda_{k,-} \sim \frac{k^{s+1} Z^\chi_k}{\lambda_{k,-}} \frac{s}{4\pi}\left(1 - \frac{\eta_\chi}{2+s}\right) cos^2(\theta_M)$ when $s < s^\star$, and $\partial_k \lambda_{k,+} \sim 0$ when $s > s^\star$.

- Furthermore, when $g = 0$ precisely, that is, when the long- and short-range nodes are completely decoupled, one obtains Eq. (1.16) exactly; such that one indentifies $\lambda_{k,+} = \Gamma^R_{k,\phi\phi}$ and $\lambda_{k,-} = \Gamma^R_{k,\chi\chi}$,

and those are exact inequalities.

$$\lambda_{k,\pm} = \left(\frac{\Gamma^R_{k,\phi\phi} \pm \Gamma^R_{k,\phi\phi}}{2} \right) + \left(\frac{\Gamma^R_{k,\chi\chi} \mp \Gamma^R_{k,\chi\chi}}{2} \right) \tag{1.16}$$

- Those aforementioned points in combination resolve the second part of the purported inconsistency; namely insofar as now the flow equations also have well-defined dimensional scalings, whereas previously they were composed of linear combinations of terms with different scalings. This discussion furthermore allows the justification of the following assessment: that when $s \neq s^\star$, either $s > s^\star$ such that $\lambda_{k,+} \to \Gamma^R_{k,\phi\phi}$ and $\lambda_{k,-} \to 0$, and one identifies the nodes ψ_+ with the short-range field ϕ and the node ψ_- disappears; or $s < s^\star$ such that $\lambda_{k,-} \to \Gamma^R_{k,\chi\chi}$ and $\lambda_{k,+} \to 0$ and one identifies the node ψ_- with the long-range field χ and the node ψ_+ disappears. This allows the identification of ψ_+ as the "short-range-like" mode and ψ_- as the "long-range-like" mode. In this connection the Behan construction Eq. (2.1) for describing the theory behaviour in the vicinity of $s \lesssim s^\star$, in essence consists of considering a theory described by a "long-range-like" mode ψ_- weakly mixed with a long-range-like mode ψ_+.

- On the other hand, at the Sak Point $s = s^\star$ the most interesting insight is obtained. As previously averred, the purpoted inconsistency pointed out in the first bullet-point above can be resolved, either (i) if one demands one of the dimensional dependancies must vanish; or (ii) if one demands that both dependancies attain the same dimensional scaling. The latter is precisely what happens at the Sak Point $s = s^\star$. Exclusively at this point, both nodes ψ_+ and ψ_- coexist and are well-defined.

- By considering the RG flow of the mixing angle, Eq. (1.14), if one denotes $\left(\Gamma^R_{k,\phi\phi} - \Gamma^R_{k,\chi\chi}\right) = D$ with a scaling Δ_D; and g_k with a scaling Δ_g (assuming that those scaling dimensions are indeed well-defined; which is the case at $s^\star$, as already discussed); then one obtains Eq. (1.17) below, such that the condition for $\partial_k \theta_M = 0$ is given by $\Delta_g = \Delta_D$, that is, precisely by $s^\star = 2 - \eta_\phi$. This by extension allows us to conclude that $\frac{2g_k}{\Gamma^R_{k,\phi\phi} - \Gamma^R_{k,\chi\chi}} = tan\left(\theta^\star_M\right)$ is constant at the cross-over $s^\star = 2 - \eta_\phi$; such that the nodes ψ_+ and ψ_- are well-defined by fixed mixtures of ϕ and χ.

$$\partial_k \theta_M = \frac{g_k D \cdot (\Delta_g - \Delta_D)}{D^2 - 4g_k^2} \tag{1.17}$$

- If we one instead considers the RG flow of the mixing angle away from $s^\star$, in that case (in the IR) $D = (\Delta_g - \Delta_D) \neq 0$ always, such that the system never attains constant θ_M and is therefore not stable. On the other hand, define $s = s^\star + \delta$ such that $s > s^\star$ corresponds to $\delta > 0$ and $s < s^\star$ corresponds to $\delta < 0$, in this case the scalings will follow Eq. (1.18), and one sees that in the IR limit $k \to 0$, although $\partial_k \theta_M$ never vanishes, it does get suppressed as a power law $k^{\left|\frac{\delta}{2}\right|}$, such that it asymptotically vanishes. This is in agreement with the previous identification of ψ_+ and ψ_- as the

corresponding short- and long-range nodes respectively at $s > s^\star$ and $s < s^\star$; with the corresponding counterpart vanishing. Crucially, one avers two distinguishable cases: that either the flow of the mixing angle $\partial_k \theta_M \to 0$ vanishes only asymptotically, corresponding to the IR limit, that is, to fixed points; or $\partial_k \theta_M = 0$ vanishes exactly for a real fixed point $\theta_M^\star$, which correpsonds to the classification of the cross-over, that is, to the Sak point.

$$\partial_k \theta_M \sim \frac{k^{\frac{\delta}{2}} \left(\tilde{\Gamma}^R_{k,\phi\phi} - k^\delta \tilde{\Gamma}^R_{k,\chi\chi} \right) \tilde{g}_k \times D}{\left(\tilde{\Gamma}^R_{k,\phi\phi} \right)^2 + k^{2\delta} \left[2\tilde{\Gamma}^R_{k,\phi\phi} \tilde{\Gamma}^R_{k,\chi\chi} + \left(\tilde{\Gamma}^R_{k,\chi\chi} \right)^2 \right] - k^\delta \left(4\tilde{g}_k^2 \right)} \tag{1.18}$$

- One concludes therefore that $\psi_\pm$ do become physical oscilating nodes either (i) at the crossover $s = s^\star$, or (ii) in the asymptotic (IR) limit of $k \to 0$ outside of the crossover region, $s \neq s^\star$. This analysis thus clarifies the mechanism of the crossover: (a) it is continuous and characterized by the (unique) point in the RG flow where the mixing angle becomes a non-asymptotic fixed point and by extension both nodes $\psi_\pm$ become "activated"; (b) namely, it is continuous because it is reachable by the continuous RG flow from regions outside of the crossover; that is to say, the aforementioned formualtion explicit derives differential equations that continously reach (and cross) the cross-over; (d) the cross-over is characterized by both modes $\psi_\pm$ becoming (d.i) non-zero fixed (by $\theta_M^\star$) mixtures of the short- and long-ranged nodes ϕ and χ; and (d.ii) namely by ψ_+ and ψ_- gaining non-zero dynamics simultaneously; (e) outside of the crossover the nodes $\psi_\pm$ are physical objects only asymptotically, and one of the nodes absolutely dominates whereas the other one node is suppressed; (f) the crossover corresponds to the emergence of a scale-invariant mixing between short- and long-range degrees of freedom; (e) the statement of a crossover is equivalent to the statement of the non-asymptotic mixing-angle stabilization; and (g) the Sak point is the point characterized such that the RG flow of the mixing-angle vanishes non-asymptotically.

- The physical operators $\psi_\pm$ are eigenvectors of the RG flow; and the crossover corresponds to the rotation of the eigenbasis. $\psi_\pm$ are furthermore scale-dependant linear operators that become scaling operators at the fixed point. Away from $s^\star$, one of the modes becomes dynamically frozen, and the RG flow projects to a single effective degree of freedom. The RG flow selects the universality class (between long and short-ranged) by suppressing one of the RG eigenmodes $\psi_\pm$; and the crossover is the selection mechanism.

- The cross-over is driven by two kinetic operators whose scaling dimension degenerates exclusively on the cross-over point, and the degeneration of scaling dimension enables operator mixing. The interaction term g_k furthermore induces an avoided-crossing of the RG eigenoperators at the crossover, such that

the cross-over is characterized by a hybridization of competing scaling operators as opposed to a simple switch in dominance. The mixing angle is in this case an order parameter for operator hybridization.

- The distinction between the flow of the mixing angle θ_M in one of the scenarios vanishing asympthotically versus on the other scenario vanishing due to the degenerancy of scaling dimension characterizes the distinction between the system on the one hand going through the cross-over and on the other hand experiencing a critical phenomena. The Sak justification emerges naturally in this construction, because the Sak point characterizes the non-asympthotic vanishing of the RG flow for the mixing-angle.

- The avoided-crossing induced by g_k introduces a mixing of competing scalings. The gap forces the two modes to hybridize and then split which leads to a smooth cross-over as opposed to a sharp transition. In the absence of a gap, $g_k = 0$, one would observe an abrupt switch from short to long range regimes. The avoided crossing thus allows θ_M to rotate smoothly, such that the cross-over occurs continuously. In particular, one avers that $g_k \neq 0$ appears to be a necessary condition for the cross-over to be continuous.

2 Appendicies Concerning LRMMs

2.1 Appendix — Solution to the Wetterich Equation

One considers the Action term specified by Eq. (2.1) below, where "..." denotes (potentially) additional self-interactions, ϕ is identified with a primary field of a CFT and χ is the fluctuation field of the long-range sector. This correpsonds to the Behan view of the cross-over.

$$S[\phi,\chi] = \int \partial^\mu \phi(x)\,\partial_\mu \phi(x)\,dx + \int \frac{(\chi(x)-\chi(y))^2}{|x-y|^{s+d}}dxdy + g\int \phi(x)\,\chi(x)\,dx + \dots \tag{2.1}$$

One projects the Wetterich Equation onto a particular LPA–prime Ansatz for the running effective action specified by Eq. (2.2) below, where $U_k[\phi,\chi]$ is the potential term correpsonding to (potentially) additional self-interactions of ϕ and of χ.

$$\Gamma_k[\phi,\chi] = \frac{1}{2}Z_k^\phi p^2\phi^2 + \frac{1}{2}Z_k^\chi p^s\chi^2 + U_k[\phi,\chi] \tag{2.2}$$

The anomalous dimensions are given by Eq. (2.3) below. In practice, one eventually removes the anomalous dimension in the long range sector, $\eta_\chi = 0$; first because it is a generally well-established fact in literature that the long-range sector will never acquire an anomalous dimension; second because this is a statement that will be argued for explicitly later-on in the IR limit.

$$\eta_\phi = -\frac{1}{Z_k^\phi}\frac{\partial Z_k^\phi}{\partial log\,(k)}\ ; \qquad \eta_\chi = -\frac{1}{Z_k^\chi}\frac{\partial Z_k^\chi}{\partial log\,(k)} \tag{2.3}$$

The (running) inverse propagator $\Gamma_k^{(2)}$ and its corresponding regularized version P_k^{-1}, are given explicitly by Eq. (2.4) below, where one in addition specifies a diagonal flow for the regulator, $\partial_k R_k$.

$$\Gamma_k^{(2)} = \begin{pmatrix} \Gamma_{k,\phi\phi} & \Gamma_{k,\phi\chi} \\ \Gamma_{k,\chi\phi} & \Gamma_{k,\chi\chi} \end{pmatrix}\ ; \quad P_k^{-1} = \begin{pmatrix} \Gamma_{k,\phi\phi} + R_{k,\phi} & \Gamma_{k,\phi\chi} \\ \Gamma_{k,\chi\phi} & \Gamma_{k,\chi\chi} + R_{k,\chi} \end{pmatrix}\ ; \quad \partial_k R_k = \begin{pmatrix} \partial_k R_{k,\phi} & 0 \\ 0 & \partial_k R_{k,\chi} \end{pmatrix} \tag{2.4}$$

Using Litim regulators, one obtains the quantities Eqs. (2.5, 2.6).

$$\Gamma_{k,\phi\phi}^R = \Gamma_{k,\phi\phi} + R_{k,\phi} = Z_k^\phi k^2 + U_{k,\phi\phi}\ ; \qquad R_{k,\phi} = Z_k^\phi\left(k^2 - p^2\right)\Theta\left(k^2 - p^2\right) \tag{2.5}$$

$$\Gamma_{k,\chi\chi}^R = \Gamma_{k,\chi\chi} + R_{k,\chi} = Z_k^\chi k^s + U_{k,\chi\chi}\ ; \qquad R_{k,\chi} = Z_k^\chi\left(k^s - p^s\right)\Theta\left(k^s - p^s\right) \tag{2.6}$$

With this setup, the Wetterich Equation is given by Eq. (2.7), where P_k denotes the regularized running propagator.

$$\partial_k\Gamma_k\left[\phi,\chi\right] = Tr\left(\left(\partial_k R_k\right)P_k\right)\ ; \qquad P_k = \frac{1}{\Gamma_{k,\phi\phi}^R\Gamma_{k,\chi\chi}^R - g_k^2}\begin{pmatrix} \Gamma_{k,\chi\chi}^R & -g_k \\ -g_k & \Gamma_{k,\phi\phi}^R \end{pmatrix}\ ; \tag{2.7}$$

The Wetterich Equation can be expanded explicitly, Eq. (2.8).

$$\partial_k\Gamma_k\left[\phi,\chi\right] = \int \frac{\left(\partial_k R_{k,\phi}\right)\Gamma_{k,\chi\chi}^R + \left(\partial_k R_{k,\chi}\right)\Gamma_{k,\phi\phi}^R}{\Gamma_{k,\phi\phi}^R\Gamma_{k,\chi\chi}^R - g_k^2}\frac{d^2\vec{p}}{(2\pi)^2} \tag{2.8}$$

2.2 Appendix — Flow of the Potential

Since the regularized running propagators $\Gamma_{k,\phi\phi}^R$ and $\Gamma_{k,\chi\chi}^R$ are regularized by such a choice, that they become explicitly independent of p; and since g_k is also explicitly independent of p; those terms can be removed from the integral; such that the integrals depend exclusively on the regulators, Eqs. (2.9, 2.10).

$$\int \left(\partial_k R_{k,\phi}\right) \frac{d^2\vec{p}}{(2\pi)^2} = k Z_k^\phi \left\{ (2-\eta_\phi) \left(\int^{p^2=k^2} \frac{d^2\vec{p}}{(2\pi)^2} \right) + \eta_\phi k^{-2} \left(\int^{p^2=k^2} p^2 \frac{d^2\vec{p}}{(2\pi)^2} \right) \right\} \tag{2.9}$$

$$\int \left(\partial_k R_{k,\chi}\right) \frac{d^2\vec{p}}{(2\pi)^2} = k^{s+1} Z_k^\chi \left\{ (2-\eta_\chi) \left(\int^{p^2=k^2} \frac{d^2\vec{p}}{(2\pi)^2} \right) + \eta_\chi k^{-s} \left(\int^{p^2=k^2} p^s \frac{d^2\vec{p}}{(2\pi)^2} \right) \right\} \tag{2.10}$$

Those integrals have explicit solutions in hands of Eq. (2.11).

$$\int^{p^2=k^2} \frac{d^2\vec{p}}{(2\pi)^2} = \frac{k^2}{4\pi} \ ; \qquad \int^{p^2=k^2} p^2 \frac{d^2\vec{p}}{(2\pi)^2} = \frac{k^4}{8\pi} \ ; \qquad \int^{p^2=k^2} p^s \frac{d^2\vec{p}}{(2\pi)^2} = \frac{1}{s+2} \frac{k^{s+2}}{2\pi} \tag{2.11}$$

Namely, one obtains Eqs. (2.12, 2.13).

$$\int \left(\partial_k R_{k,\phi}\right) \frac{d^2\vec{p}}{(2\pi)^2} = k^3 Z_k^\phi \frac{1}{2\pi} \left(1 - \frac{\eta_\phi}{4}\right) = k^3 Z_k^\phi C_\phi \tag{2.12}$$

$$\int \left(\partial_k R_{k,\chi}\right) \frac{d^2\vec{p}}{(2\pi)^2} = k^{s+3} Z_k^\chi \frac{s}{2\pi} \left(1 - \frac{\eta_\chi}{s+2}\right) = k^{s+3} Z_k^\chi C_\chi \tag{2.13}$$

Furthermore, the running inverse propagators can be written in dimensionless form, Eq. (2.14).

$$\Gamma^R_{k,\phi\phi} = k^2 Z_k^\phi \left(1 + \tilde{U}_{k,\phi\phi}\right) = k^2 Z_k^\phi \tilde{\Gamma}^R_{k,\phi\phi} \ ; \qquad \Gamma^R_{k,\chi\chi} = k^s Z_k^\chi \left(1 + \tilde{U}_{k,\chi\chi}\right) = k^s Z_k^\chi \tilde{\Gamma}^R_{k,\chi\chi} \tag{2.14}$$

This is justified namely by the adimensional quantities, Eq. (2.15).

$$U_k = k^2 \tilde{U}_k \ ; \qquad \phi = k^{\frac{\eta_\phi}{2}} \tilde{\phi} \ ; \qquad \chi = k^{\frac{d-s+\eta_\chi}{2}} \tilde{\chi} \tag{2.15}$$

Combining all those terms, Eqs. (2.12, 2.13, 2.14) onto Eq. (2.8), we reduce the right-handed side of Eq. (2.8). to an adimensionless quantity, Eq. (2.16).

$$\partial_k \Gamma_k \left[\phi, \chi\right] = \left(\frac{k^{s+3} Z_k^\phi Z_k^\chi}{k^{s+2} Z_k^\phi Z_k^\chi} \right) \frac{C_\phi \tilde{\Gamma}^R_{k,\chi\chi} + C_\chi \tilde{\Gamma}^R_{k,\phi\phi}}{\tilde{\Gamma}^R_{k,\phi\phi} \tilde{\Gamma}^R_{k,\chi\chi} - \tilde{g}_k^2} = k \frac{C_\phi \tilde{\Gamma}^R_{k,\chi\chi} + C_\chi \tilde{\Gamma}^R_{k,\phi\phi}}{\tilde{\Gamma}^R_{k,\phi\phi} \tilde{\Gamma}^R_{k,\chi\chi} - \tilde{g}_k^2} \tag{2.16}$$

The left-handed side of the Wetterich Equation, Eq. (2.8), can similarly be written in dimensionless form, Eq. (2.17). Furthermore, one replaces $\Gamma_k \left[\phi, \chi\right] \to U_k \left[\phi, \chi\right]$ by considering the expression at zero-momentum

evaluation.

$$k^{-1}\partial_k U_k = k^{-1}\partial_k\left(k^2\tilde{U}_k\right) = 2\tilde{U}_t + k\left(\partial_k\tilde{\phi}\right)\partial_{\tilde{\phi}}\tilde{U}_t + k\left(\partial_k\tilde{\chi}\right)\partial_{\tilde{\chi}}\tilde{U}_t + \partial_t\tilde{U}_t = 2\tilde{U}_t - \frac{\eta_\phi}{2}\tilde{U}_{t,\phi} - \frac{d-s+\eta_\chi}{2}\tilde{U}_{t,\chi} + \partial_t\tilde{U}_t \tag{2.17}$$

The Wetterich Equation for the potential thus take explicit form, Eq. (2.18), remembering the definitions, $C_\phi = \frac{1}{2\pi}\left(1-\frac{\eta_\phi}{4}\right)$ and $C_\chi = \frac{s}{2\pi}\left(1-\frac{\eta_\chi}{s+2}\right)$

$$\partial_t\tilde{U}_t = -2\tilde{U}_t + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}_{t,\phi} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}_{t,\chi} + \frac{C_\chi\left(1+\tilde{U}_{t,\phi\phi}\right) + C_\phi\left(1+\tilde{U}_{t,\chi\chi}\right)}{\left(1+\tilde{U}_{t,\phi\phi}\right)\left(1+\tilde{U}_{t,\chi\chi}\right) - g^2} \tag{2.18}$$

2.3 Solution in a Diagonalized Basis

From section 2.1 Appendix — Solution to the Wetterich Equation, Eqs. (2.7, 2.8), a potential alternative step would consist of diagonalizing the propagator onto two linearly-independent basis. Namely, one considers the eigenproblem specified by Eq. (2.19) for the diagonalization of the running inverse propagator P_k^{-1}.

$$P_k^{-1}\vec{v} = \lambda_k\vec{v}\ ;\ P_k^{-1} = MD^{-1}M^{-1}\ ;\ D^{-1} = diag\begin{pmatrix} \frac{1}{\lambda_{k,+}} & \frac{1}{\lambda_{k,-}} \end{pmatrix}\ ;\ M = \begin{pmatrix} \vec{v}_+ & \vec{v}_- \end{pmatrix} \tag{2.19}$$

The Eigenvalues are obtained from solving the quadratic $\lambda^2 - \left(\Gamma^R_{k,\phi\phi} + \Gamma^R_{k,\chi\chi}\right)\lambda + \left(\Gamma^R_{k,\phi\phi}\Gamma^R_{k,\chi\chi} - g_k^2\right) = 0$, Eq. (2.20).

$$\lambda_{k,\pm} = \left(\frac{\Gamma^R_{k,\phi\phi}+\Gamma^R_{k,\chi\chi}}{2}\right) \pm \sqrt{\left(\frac{\Gamma^R_{k,\phi\phi}-\Gamma^R_{k,\chi\chi}}{2}\right)^2 + g_k^2} = \left(\frac{\Gamma^R_{k,\phi\phi}+\Gamma^R_{k,\chi\chi}}{2}\right) \pm \left(\frac{\Gamma^R_{k,\phi\phi}-\Gamma^R_{k,\chi\chi}}{2}\right)\sqrt{1+\left(\frac{2g_k}{\Gamma^R_{k,\phi\phi}-\Gamma^R_{k,\chi\chi}}\right)^2} \tag{2.20}$$

The corresponding eigenvectors satisfy $\vec{v}_\pm = \begin{pmatrix} g_k \\ \lambda_{k,\pm} - \Gamma^R_{k,\phi\phi} \end{pmatrix}$, such that we build matrices, Eq. (2.21).

$$M = \begin{pmatrix} \vec{v}_+ & \vec{v}_- \end{pmatrix} = \begin{pmatrix} g_k & g_k \\ \lambda_{k,+} - \Gamma^R_{k,\phi\phi} & \lambda_{k,-} - \Gamma^R_{k,\phi\phi} \end{pmatrix}\ ;\quad M^{-1} = \frac{1}{g_k\left(\lambda_{k,+}-\lambda_{k,-}\right)}\begin{pmatrix} \Gamma^R_{k,\phi\phi} - \lambda_{k,-} & g_k \\ \lambda_{k,+} - \Gamma^R_{k,\phi\phi} & -g_k \end{pmatrix} \tag{2.21}$$

Those matrices can be expressed in terms of a rotation matrix by means of the transformations Eq. (2.22).

$$tan\left(2\theta_M\right) = \frac{2g_k}{\Gamma^R_{k,\phi\phi} - \Gamma^R_{k,\chi\chi}} \;; \qquad cos^2\left(\theta_M\right) = \frac{\Gamma^R_{k,\phi\phi} - \lambda_{k,-}}{\lambda_{k,+} - \lambda_{k,-}} \;; \qquad sin^2\left(\theta_M\right) = \frac{\lambda_{k,+} - \Gamma^R_{k,\phi\phi}}{\lambda_{k,+} - \lambda_{k,-}} \;; \tag{2.22}$$

The Wetterich Equation can thus be written in a diagonalized form, Eq. (2.23), where exploits the cyclicity of the trace to move the affect M and M^{-1} on the regulator $\partial_k R_k$.

$$\partial_k \Gamma_k\left[\phi,\chi\right] = Tr\left(\left(\partial_k R_k\right) P_k\right) = Tr\left(\left(\partial_k R_k\right) M D^{-1} M^{-1}\right) = Tr\left(\left\{M^{-1}\left(\partial_k R_k\right) M\right\} D^{-1}\right) = Tr\left(\left(\partial_k \tilde{R}_k\right) D^{-1}\right) \tag{2.23}$$

In this step, one emphasizes, crucially, that although the regulator is in general an arbitrary function that can be chosen at will; in this particular case the regulators were already explicitly affixated by Eqs. (2.5, 2.6), so as to remove the p-dependance on $\lambda_\pm$ and on the regulated terms $\Gamma^R_{k,\phi\phi}$ and $\Gamma^R_{k,\chi\chi}$. This affixation in essence allows analytically-integrable, closed-form flow equations for the diagonalized nodes. On the converse side, this leads thereto, that the regulator will need to be explicitly rotated to the new basis $\partial_k \tilde{R}_k = M^{-1}\left(\partial_k R_k\right) M$.

In the rotated basis, the explicit computation of the trace elements of $\partial_k \tilde{R}_k$ is given by Eqs. (2.24, 2.25) below. Since $D^{-1} = diag\begin{pmatrix} \frac{1}{\lambda_{k,+}} & \frac{1}{\lambda_{k,-}} \end{pmatrix}$ is already diagonal, it contributes to the trace simply by multiplying the $\left(\partial_k \tilde{R}_k\right)_{11}$ and $\left(\partial_k \tilde{R}_k\right)_{22}$ elements; so they are for now not a point of concern.

$$\left(\partial_k \tilde{R}_k\right)_{11} = \partial_k R_{k,+} = cos^2\left(\theta_M\right)\partial_k R_{k,\phi} + sin^2\left(\theta_M\right)\partial_k R_{k,\chi} = \frac{\Gamma^R_{k,\phi\phi} - \lambda_{k,-}}{\lambda_{k,+} - \lambda_{k,-}}\partial_k R_{k,\phi} + \frac{\lambda_{k,+} - \Gamma^R_{k,\phi\phi}}{\lambda_{k,+} - \lambda_{k,-}}\partial_k R_{k,\chi} \tag{2.24}$$

$$\left(\partial_k \tilde{R}_k\right)_{22} = \partial_k R_{k,-} = sin^2\left(\theta_M\right)\partial_k R_{k,\phi} + cos^2\left(\theta_M\right)\partial_k R_{k,\chi} = \frac{\lambda_{k,+} - \Gamma^R_{k,\phi\phi}}{\lambda_{k,+} - \lambda_{k,-}}\partial_k R_{k,\phi} + \frac{\Gamma^R_{k,\phi\phi} - \lambda_{k,-}}{\lambda_{k,+} - \lambda_{k,-}}\partial_k R_{k,\chi} \tag{2.25}$$

The Wetterich equation is thus given by Eq. (2.26).

$$\begin{aligned}\partial_k \Gamma_k\left[\phi,\chi\right] &= \partial_k \Gamma_{k,\psi_+}\left[\phi,\chi\right] + \partial_k \Gamma_{k,\psi_-}\left[\phi,\chi\right] = \int\left(\frac{\partial_k R_{k,+}}{\lambda_{k,+}} + \frac{\partial_k R_{k,-}}{\lambda_{k,-}}\right)dp = \\ &= \int\left(\frac{cos^2\left(\theta_M\right)\partial_k R_{k,\phi} + sin^2\left(\theta_M\right)\partial_k R_{k,\chi}}{\lambda_{k,+}} + \frac{sin^2\left(\theta_M\right)\partial_k R_{k,\phi} + cos^2\left(\theta_M\right)\partial_k R_{k,\chi}}{\lambda_{k,-}}\right)dp\end{aligned} \tag{2.26}$$

Here, one observes that the Wetterich equation described in Eq. (2.26) corresponds to the equation that

would be obtained by taking the diagonalized action Eq. (2.27) as a starting point. In other words, this procedure shows that (at least in principle) the proposed Behan action Eq. (2.1) could be written in terms of two "synthetic", diagonalized oscillating nodes whose "inverse propagators" are given by the eigenvalues $\lambda_{\pm}$; and one furthermore identifies $\Gamma_{k,\psi_{\pm}} = \lambda_{k,\pm}$.

$$S\left[\psi_{+},\psi_{-}\right]=\int\left(\lambda_{k,+}\psi_{+}^{2}+\lambda_{k,-}\psi_{-}^{2}\right)dp \tag{2.27}$$

In the subsequent this construction will be discussed in more details; as well as what $\psi_{\pm}$ are meant to be understood as; and whether they are even real physical objects to begin with. First of all, as a consistency check, one notices that by using $(\lambda_{k,+}+\lambda_{k,-})=\left(\Gamma_{k,\phi\phi}^{R}+\Gamma_{k,\chi\chi}^{R}\right)$ and $\lambda_{k,+}\lambda_{k,-}=\Gamma_{k,\phi\phi}^{R}\Gamma_{k,\chi\chi}^{R}-g^{2}$, and by explicitly expanding $\partial_k R_{k,+}$ and $\partial_k R_{k,-}$ in terms of the eigenvectors and propagators, one obtains Eqs. (2.28, 2.29).

$$\left(\frac{1}{\lambda_{k,+}}\frac{\Gamma_{k,\phi\phi}^{R}-\lambda_{k,-}}{\lambda_{k,+}-\lambda_{k,-}}+\frac{1}{\lambda_{k,-}}\frac{\lambda_{k,+}-\Gamma_{k,\phi\phi}^{R}}{\lambda_{k,+}-\lambda_{k,-}}\right)\partial_k R_{k,\phi}=\frac{\left(\lambda_{+}^{2}-\lambda_{-}^{2}\right)-\Gamma_{k,\phi\phi}^{R}\left(\lambda_{k,+}-\lambda_{k,-}\right)}{\lambda_{k,+}\lambda_{k,-}\left(\lambda_{k,+}-\lambda_{k,-}\right)}\partial_k R_{k,\phi}=\frac{\Gamma_{k,\chi\chi}^{R}}{\lambda_{k,+}\lambda_{k,-}}\partial_k R_{k,\phi} \tag{2.28}$$

$$\left(\frac{1}{\lambda_{k,+}}\frac{\lambda_{k,+}-\Gamma_{k,\phi\phi}^{R}}{\lambda_{k,+}-\lambda_{k,-}}+\frac{1}{\lambda_{k,-}}\frac{\Gamma_{k,\phi\phi}^{R}-\lambda_{k,-}}{\lambda_{k,+}-\lambda_{k,-}}\right)\partial_k R_{k,\chi}=\frac{\Gamma_{k,\phi\phi}^{R}\left(\lambda_{k,+}-\lambda_{k,-}\right)}{\lambda_{k,+}\lambda_{k,-}\left(\lambda_{k,+}-\lambda_{k,-}\right)}\partial_k R_{k,\chi}=\frac{\Gamma_{k,\phi\phi}^{R}}{\lambda_{k,+}\lambda_{k,-}}\partial_k R_{k,\chi} \tag{2.29}$$

Such that Eq. (2.26) can be reduced to Eq. (2.30), that is, to the previously-derived result in section 2.1 Appendix — Solution to the Wetterich Equation, Eq. (2.8), corresponding to the undiagonalized basis.

$$\partial_k\Gamma_k=\int\left(\frac{\partial_k R_{k,+}}{\lambda_{k,+}}+\frac{\partial_k R_{k,-}}{\lambda_{k,-}}\right)dp=\int\frac{\left(\partial_k R_{k,\phi}\right)\Gamma_{k,\chi\chi}^{R}+\left(\partial_k R_{k,\chi}\right)\Gamma_{k,\phi\phi}^{R}}{\Gamma_{k,\phi\phi}^{R}\Gamma_{k,\chi\chi}^{R}-g^{2}}dp \tag{2.30}$$

Returning to Eq. (2.26), one notices that the dependance on p lies (again) only on the regulators $\partial_k R_{k,\phi}$ and $\partial_k R_{k,\chi}$. The explicit solution of the integral dependencies was already solved in section 2.1 Appendix — Solution to the Wetterich Equation, Eqs. (2.12, 2.13). The corresponding flows of the renormalization flux are given by Eqs. (2.31, 2.32).

$$\partial_k\lambda_{k,+}=\frac{k^{3-\eta_\phi}}{\lambda_{k,+}}\frac{1}{4\pi}\left(1-\frac{\eta_\phi}{4}\right)cos^{2}\left(\theta_M\right)+\frac{k^{s+1}Z_k^{\chi}}{\lambda_{k,+}}\frac{s}{4\pi}\left(1-\frac{\eta_\chi}{2+s}\right)sin^{2}\left(\theta_M\right) \tag{2.31}$$

$$\partial_k\lambda_{k,-}=\frac{k^{s+1}Z_k^{\chi}}{\lambda_{k,-}}\frac{s}{4\pi}\left(1-\frac{\eta_\chi}{2+s}\right)cos^{2}\left(\theta_M\right)+\frac{k^{3-\eta_\phi}}{\lambda_{k,-}}\frac{1}{4\pi}\left(1-\frac{\eta_\phi}{4}\right)sin^{2}\left(\theta_M\right) \tag{2.32}$$

One furthermore finds the flow of θ_M explicitly by taking the derivative with respect to Eq. (2.22), Eq. (2.33).

$$\partial_k \theta_M = \frac{\left(\Gamma^R_{k,\phi\phi} - \Gamma^R_{k,\chi\chi}\right) \cdot \partial_k g_k - g_k \cdot \partial_k \left(\Gamma^R_{k,\phi\phi} - \Gamma^R_{k,\chi\chi}\right)}{\left(\Gamma^R_{k,\phi\phi} - \Gamma^R_{k,\chi\chi}\right)^2 - 4g_k^2} \tag{2.33}$$

As a final algebraic step that will be relevant in subsequent sections, one considers the taylor-expand $\lambda_{k,\pm}$ at small g, Eq. (2.34).

$$\lambda_{k,\pm} \approx \left(\frac{\Gamma^R_{k,\phi\phi} \pm \Gamma^R_{k,\phi\phi}}{2}\right) + \left(\frac{\Gamma^R_{k,\chi\chi} \mp \Gamma^R_{k,\chi\chi}}{2}\right) \pm \frac{g_k^2}{\Gamma^R_{k,\phi\phi} - \Gamma^R_{k,\chi\chi}} + \mathcal{O}\left(g^4\right) \tag{2.34}$$

2.4 The RG Flow Equations

In order to explicitly derive the flows from Eq. Eq. (2.18), on considers for example the potential term specified by Eq. (2.35) below, where "..." denotes (potentially) additional interaction terms. For simplicity, one considers as a start truncations at $\mathcal{O}\left(\phi^4\right)$ and $\mathcal{O}\left(\chi^4\right)$ interactions and $g_t\phi\chi$ as the only mixing term.

$$U_t\left[\phi,\chi\right] = m^2_{\phi,t}\phi^2 + \frac{\lambda_{\phi,t}}{4!}\phi^4 + m^2_{\chi,t}\chi^2 + \frac{\lambda_{\chi,t}}{4!}\chi^4 + g_t\phi\chi + \ldots \tag{2.35}$$

For the remaining of the section, one furthermore considers the following notation: $\tilde{U}^{(m,n)}_{t,\phi\chi}$ denotes the nth derivative with respect to χ of the mth derivative with respect to ϕ of $\tilde{U}_t$. $\tilde{U}^{(m)}_{t,\chi}$ or $\tilde{U}^{(m)}_{t,\phi}$ are the mth derivatives of $\tilde{U}_t$with respect to χ or ϕ respectively and so on. As an example to clarify, one owuld denote $\tilde{U}^{(2)}_{t,\chi} = \tilde{U}_{t,\chi\chi} = \tilde{U}^{(0,2)}_{t,\phi\chi}$ the second derivative of $\tilde{U}_t$with respect to χ; or $\tilde{U}^{(2,1)}_{t,\phi\chi} = \tilde{U}_{t,\phi\phi\chi}$ the derivative of $\tilde{U}_{t,\phi\phi}$ with respect to χ; or $\tilde{U}^{(1,1)}_{t,\phi\chi} = \tilde{U}_{t,\phi\chi}$ the derivative of $\tilde{U}_t$with respect to χ and ϕ simultaneously, etc. With this particular notation, the Wetterich Equation, Eq. (2.18), is written as Eq. (2.36).

$$\partial_t \tilde{U}^{(0,0)}_{t,\phi\chi} = -2\tilde{U}^{(0,0)}_{t,\phi\chi} + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}^{(1,0)}_{t,\phi\chi} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}^{(0,1)}_{t,\phi\chi} + \frac{C_\chi\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right) + C_\phi\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)}{\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right) - g^2} \tag{2.36}$$

To simplify the problem, one re-writes one considers Eq. (2.37 – 2.41)

$$\partial_t \tilde{U}^{(0,0)}_{t,\phi\chi} = \mathcal{D}^{(0,0)}_{\phi\chi} + \mathcal{P}^{(0,0)}_{\phi\chi} \tag{2.37}$$

$$\mathcal{D}^{(0,0)}_{\phi\chi} = -2\tilde{U}^{(0,0)}_{t,\phi\chi} + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}^{(1,0)}_{t,\phi\chi} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}^{(0,1)}_{t,\phi\chi} \tag{2.38}$$

$$\mathcal{P}^{(0,0)}_{\phi\chi} = \mathcal{A}^{(0,0)}_{\phi\chi}\left(\mathcal{B}^{(0,0)}_{\phi\chi}\right)^{-1} \tag{2.39}$$

$$\mathcal{A}^{(0,0)}_{\phi\chi} = C_\chi\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right) + C_\phi\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right) \tag{2.40}$$

$$\mathcal{B}^{(0,0)}_{\phi\chi} = \left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right) - g^2 \tag{2.41}$$

Up to the truncation choice, the derivatives with respect to $\mathcal{A}^{(0,0)}_{\phi\chi}$ are written by Eqs. (2.42, 2.43)

$$\mathcal{A}^{(1,0)}_{\phi\chi} = C_\chi\tilde{U}^{(3,0)}_{t,\phi\chi}\ ; \qquad \mathcal{A}^{(0,1)}_{\tilde{\phi}\tilde{\psi}} = C_\phi\tilde{U}^{(0,3)}_{t,\phi\chi} \tag{2.42}$$

$$\mathcal{A}^{(2,0)}_{\phi\chi} = C_\chi\tilde{U}^{(4,0)}_{t,\phi\chi}\ ; \qquad \mathcal{A}^{(0,2)}_{\tilde{\phi}\tilde{\psi}} = C_\phi\tilde{U}^{(0,4)}_{t,\phi\chi} \tag{2.43}$$

Up to the truncation choice, the derivatives with respect to $\mathcal{B}^{(0,0)}_{\phi\chi}$ are given by Eqs. (2.44, 2.45).

$$\mathcal{B}^{(1,0)}_{\phi\chi} = \tilde{U}^{(3,0)}_{t,\phi\chi}\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)\ ; \qquad \mathcal{B}^{(0,1)}_{\phi\chi} = \tilde{U}^{(0,3)}_{t,\phi\chi}\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right) \tag{2.44}$$

$$\mathcal{B}^{(2,0)}_{\phi\chi} = \tilde{U}^{(4,0)}_{t,\phi\chi}\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)\ ; \qquad \mathcal{B}^{(0,2)}_{\phi\chi} = \tilde{U}^{(0,4)}_{t,\phi\chi}\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right) \tag{2.45}$$

In the symmetrically-broken phase, one additionally obtains the non-vanishing term, Eq. (2.46).

$$\mathcal{B}^{(1,1)}_{\phi\chi} = \tilde{U}^{(3,0)}_{t,\phi\chi}\tilde{U}^{(0,3)}_{t,\phi\chi} \tag{2.46}$$

One computes the derivatives of $\mathcal{P}^{(0,0)}_{\phi\chi}$, Eqs. (2.47 – 2.51). In the symmetric phase, one exploits $\partial_t\tilde{U}^{(1,0)}_{t,\phi\chi} = 0$ and $\partial_t\tilde{U}^{(3,0)}_{t,\phi\chi} = 0$ to conclude that $\mathcal{P}^{(1,0)}_{\phi\chi} = 0$ and $\mathcal{P}^{(3,0)}_{\phi\chi} = 0$; and even further, that $\mathcal{B}^{(m,n)}_{\phi\chi} = \mathcal{A}^{(m,n)}_{\phi\chi} = 0$ for any m or n odd. One additionaly uses the truncation choices to note that$\mathcal{A}^{(n,0)}_{\phi\chi} = 0$ and $\mathcal{B}^{(n,0)}_{\phi\chi} = 0$ for all $n \geq 3$. More specifically, one must clarify that those conditions apply specifically at the point of evaluation, $(\cdot)|_{\phi,\chi=0}$. Finally, one notices that those expressions are symmetric with respect to $\mathcal{P}^{(n,0)}_{\phi\chi} \to \mathcal{P}^{(0,n)}_{\phi\chi}$, up to changing indicies.

$$\mathcal{P}_{\phi\chi}^{(1,0)} = \mathcal{A}_{\phi\chi}^{(1,0)} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-1} - \mathcal{A}_{\phi\chi}^{(0,0)} \mathcal{B}_{\phi\chi}^{(1,0)} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-2} \tag{2.47}$$

$$\mathcal{P}_{\phi\chi}^{(2,0)} = \mathcal{A}_{\phi\chi}^{(2,0)} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-1} - \mathcal{A}_{\phi\chi}^{(0,0)} \mathcal{B}_{\phi\chi}^{(2,0)} \left(\mathcal{B}_{\tilde{\phi}\tilde{\psi}}^{(0,0)}\right)^{-2} \tag{2.48}$$

$$\mathcal{P}_{\phi\chi}^{(3,0)} = -2\mathcal{A}_{\phi\chi}^{(2,0)} \mathcal{B}_{\phi\chi}^{(1,0)} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-2} + 2\mathcal{A}_{\phi\chi}^{(0,0)} \mathcal{B}_{\phi\chi}^{(1,0)} \mathcal{B}_{\phi\chi}^{(2,0)} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-3} \tag{2.49}$$

$$\mathcal{P}_{\phi\chi}^{(4,0)} = -2\mathcal{A}_{\phi\chi}^{(2,0)} \mathcal{B}_{\phi\chi}^{(2,0)} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-2} + 2\mathcal{A}_{\phi\chi}^{(0,0)} \left(\mathcal{B}_{\phi\chi}^{(2,0)}\right)^{2} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-3} \tag{2.50}$$

$$\begin{aligned}\mathcal{P}_{\phi\chi}^{(1,1)} = \mathcal{A}_{\phi\chi}^{(1,1)} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-1} - \mathcal{A}_{\phi\chi}^{(1,0)} \mathcal{B}_{\phi\chi}^{(0,1)} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-2} - \mathcal{A}_{\phi\chi}^{(0,1)} \mathcal{B}_{\phi\chi}^{(1,0)} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-2} \\ - \mathcal{A}_{\phi\chi}^{(0,0)} \mathcal{B}_{\phi\chi}^{(1,1)} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-2} + 2\mathcal{A}_{\phi\chi}^{(0,0)} \mathcal{B}_{\phi\chi}^{(1,0)} \mathcal{B}_{\phi\chi}^{(0,1)} \left(\mathcal{B}_{\phi\chi}^{(0,0)}\right)^{-3}\end{aligned} \tag{2.51}$$

Finally, one computes derivatives with respect to $\mathcal{D}_{\phi\chi}^{(0,0)}$, Eqs. (2.52 – 2.58).

$$\mathcal{D}_{\phi\chi}^{(0,0)} = -2\tilde{U}_{t,\phi\chi}^{(0,0)} + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}_{t,\phi\chi}^{(1,0)} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}_{t,\phi\chi}^{(0,1)} \tag{2.52}$$

$$\mathcal{D}_{\phi\chi}^{(m,n)} = -\left(2 - \frac{m}{2}\eta_\phi - \frac{n}{2}\left(2-s+\eta_\chi\right)\right)\tilde{U}_{t,\phi\chi}^{(m,n)} + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}_{t,\phi\chi}^{(m+1,n)} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}_{t,\phi\chi}^{(m,n+1)} \tag{2.53}$$

$$\mathcal{D}_{\phi\chi}^{(2,0)} = -\left(2-\eta_\phi\right)\tilde{U}_{t,\phi\chi}^{(2,0)} + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}_{t,\phi\chi}^{(3,0)} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}_{t,\phi\chi}^{(2,1)} \tag{2.54}$$

$$\mathcal{D}_{\phi\chi}^{(0,2)} = -\left(s-\eta_\chi\right)\tilde{U}_{t,\phi\chi}^{(0,2)} + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}_{t,\phi\chi}^{(1,2)} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}_{t,\phi\chi}^{(0,3)} \tag{2.55}$$

$$\mathcal{D}_{\phi\chi}^{(4,0)} = -2\left(1-\eta_\phi\right)\tilde{U}_{t,\phi\chi}^{(4,0)} + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}_{t,\phi\chi}^{(5,0)} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}_{t,\phi\chi}^{(4,1)} \tag{2.56}$$

$$\mathcal{D}_{\phi\chi}^{(0,4)} = -2\left(s-1-\eta_\chi\right)\tilde{U}_{t,\phi\chi}^{(0,4)} + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}_{t,\phi\chi}^{(4,1)} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}_{t,\phi\chi}^{(0,5)} \tag{2.57}$$

$$\mathcal{D}_{\phi\chi}^{(1,1)} = -\left(1 + \frac{s-\eta_\chi-\eta_\phi}{2}\right)\tilde{U}_{t,\phi\chi}^{(1,1)} + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}_{t,\phi\chi}^{(2,1)} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}_{t,\phi\chi}^{(1,2)} \tag{2.58}$$

Combining all terms, one obtains Eqs. (2.59 – 2.62).

$$\partial_t \tilde{U}^{(2,0)}_{t,\phi\chi} = -\left(2-\eta_\phi\right)\tilde{U}^{(2,0)}_{t,\phi\chi} + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}^{(3,0)}_{t,\phi\chi} + \frac{C_\chi \tilde{U}^{(4,0)}_{t,\phi\chi}}{\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)-g^2}$$
$$-\tilde{U}^{(4,0)}_{t,\phi\chi}\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)\frac{\left[C_\chi\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)+C_\phi\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)\right]}{\left[\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)-g^2\right]^2} \tag{2.59}$$

$$\partial_t \tilde{U}^{(4,0)}_{t,\phi\chi} = -2\left(1-\eta_\phi\right)\tilde{U}^{(4,0)}_{t,\phi\chi} - 2\frac{C_\chi\left(\tilde{U}^{(4,0)}_{t,\phi\chi}\right)^2\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)}{\left[\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)-g^2\right]^2}$$
$$+2\left(\tilde{U}^{(0,4)}_{t,\phi\chi}\right)^2\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)^2\frac{\left[C_\chi\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)+C_\phi\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)\right]}{\left[\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)-g^2\right]^3} \tag{2.60}$$

$$\partial_t \tilde{U}^{(0,2)}_{t,\phi\chi} = -\left(s-\eta_\chi\right)\tilde{U}^{(0,2)}_{t,\phi\chi} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}^{(0,3)}_{t,\phi\chi} + \frac{C_\phi \tilde{U}^{(0,4)}_{t,\phi\chi}}{\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)-g^2}$$
$$-\tilde{U}^{(0,4)}_{t,\phi\chi}\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)\frac{\left[C_\chi\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)+C_\phi\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)\right]}{\left[\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)-g^2\right]^2} \tag{2.61}$$

$$\partial_t \tilde{U}^{(0,4)}_{t,\phi\chi} = -2\left(s-1-\eta_\chi\right)\tilde{U}^{(0,4)}_{t,\phi\chi} - 2\frac{C_\phi\left(\tilde{U}^{(0,4)}_{t,\phi\chi}\right)^2\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)}{\left[\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)-g^2\right]^2}$$
$$+2\left(\tilde{U}^{(4,0)}_{t,\phi\chi}\right)^2\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)^2\frac{\left[C_\chi\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)+C_\phi\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)\right]}{\left[\left(1+\tilde{U}^{(2,0)}_{t,\phi\chi}\right)\left(1+\tilde{U}^{(0,2)}_{t,\phi\chi}\right)-g^2\right]^3} \tag{2.62}$$

In the symmetric Phase, those terms are written as Eqs. (2.63 – 2.66).

$$\partial_t \tilde{m}_\phi^2 = -\left(2-\eta_\phi\right)\tilde{m}_\phi^2 + \frac{C_\chi \tilde{\lambda}_\phi}{\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g^2} - \tilde{\lambda}_\phi\left(1+\tilde{m}_\chi^2\right)\frac{\left[C_\chi\left(1+\tilde{m}_\phi^2\right)+C_\phi\left(1+\tilde{m}_\chi^2\right)\right]}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g^2\right]^2} \tag{2.63}$$

$$\partial_t \tilde{\lambda}_\phi = -2\left(1-\eta_\phi\right)\tilde{\lambda}_\phi - 2\frac{C_\chi\left(\tilde{\lambda}_\phi\right)^2\left(1+\tilde{m}_\chi^2\right)}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g^2\right]^2} + 2\left(\tilde{\lambda}_\phi\right)^2\left(1+\tilde{m}_\chi^2\right)^2\frac{\left[C_\chi\left(1+\tilde{m}_\phi^2\right)+C_\phi\left(1+\tilde{m}_\chi^2\right)\right]}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g^2\right]^3} \tag{2.64}$$

$$\partial_t \tilde{m}_\chi^2 = -\left(s-\eta_\chi\right)\tilde{m}_\chi^2 + \frac{C_\phi \tilde{\lambda}_\chi}{\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g^2} - \tilde{\lambda}_\chi\left(1+\tilde{m}_\phi^2\right)\frac{\left[C_\chi\left(1+\tilde{m}_\phi^2\right)+C_\phi\left(1+\tilde{m}_\chi^2\right)\right]}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g^2\right]^2} \tag{2.65}$$

$$\partial_t \tilde{\lambda}_\chi = -2\left(s-1-\eta_\chi\right)\tilde{\lambda}_\chi - 2\frac{C_\phi\left(\tilde{\lambda}_\chi\right)^2\left(1+\tilde{m}_\phi^2\right)}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g^2\right]^2} + 2\left(\tilde{\lambda}_\chi\right)^2\left(1+\tilde{m}_\phi^2\right)^2\frac{\left[C_\chi\left(1+\tilde{m}_\phi^2\right)+C_\phi\left(1+\tilde{m}_\chi^2\right)\right]}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g^2\right]^3} \tag{2.66}$$

2.5 Small $g \to 0$ Limit

In the $g \to 0$ limit, one considers the taylor expansion, Eq. (2.67).

$$\frac{1}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g_k^2\right]} \approx \frac{1}{\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)}\left(1+\frac{g^2}{\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)}\right) + \mathcal{O}\left(g^4\right) \tag{2.67}$$

One uses this simplification to write, for example, Eqs. (2.68, 2.69).

$$\frac{\tilde{\lambda}_{k,\phi}\left(1+\tilde{m}_\chi^2\right)\left(1+\tilde{m}_\phi^2\right)}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g_k^2\right]^2} = \frac{\tilde{\lambda}_{k,\phi}}{\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)} + \frac{g^2\tilde{\lambda}_{k,\phi}}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)\right]^2} + \mathcal{O}\left(g^4\right) \tag{2.68}$$

$$\frac{2\left(\tilde{\lambda}_{k,\phi}\right)^2\left(1+\tilde{m}_\chi^2\right)^2\left(1+\tilde{m}_\phi^2\right)}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g_k^2\right]^3} = \frac{2\tilde{\lambda}_{k,\phi}\left(1+\tilde{m}_\chi^2\right)}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g_k^2\right]^2} + \frac{2g^2\left(\tilde{\lambda}_{k,\phi}\right)^2\left(1+\tilde{m}_\chi^2\right)}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)-g_k^2\right]^3} + \mathcal{O}\left(g^4\right) \tag{2.69}$$

The RG flow equations in the symmetric phase thus reduce to Eqs. (2.70 – 2.73). This is consistent with the expectation that as $g_k \to 0$, the two nodes should decouple decouple.

$$\partial_k \tilde{m}_\phi^2 = -\left(2-\eta_{\tilde{\rho}}\right)\tilde{m}_\phi^2 + -\frac{\tilde{\lambda}_{k,\phi}\xi_{\tilde{\phi}}}{\left(1+\tilde{m}_\phi^2\right)^2} - \frac{g^2\tilde{\lambda}_{k,\phi}}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)\right]^2} \tag{2.70}$$

$$\partial_k \tilde{\lambda}_{k,\phi} = -2\left(1-\eta_{\tilde{\rho}}\right)\tilde{\lambda}_{k,\phi} + \frac{2\xi_{\tilde{\phi}}\left(\tilde{\lambda}_{k,\phi}\right)^2}{\left(1+\tilde{m}_\phi^2\right)^3} + \frac{2g^2\left(\tilde{\lambda}_{k,\phi}\right)^2\left(1+\tilde{m}_\chi^2\right)}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)\right]^3} \tag{2.71}$$

$$\partial_k \tilde{m}_\chi^2 = -s\tilde{m}_\chi^2 + \frac{\tilde{\lambda}_{k,\chi}}{\left(1+\tilde{m}_\chi^2\right)^2} - \frac{g^2\tilde{\lambda}_{k,\chi}\xi_{\tilde{\phi}}}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)\right]^2} \tag{2.72}$$

$$\partial_k \tilde{\lambda}_{k,\chi} = -2\left(s-1\right)\tilde{\lambda}_{k,\chi} + \frac{2\left(\tilde{\lambda}_{k,\chi}\right)^2}{\left(1+\tilde{m}_\chi^2\right)^3} + \frac{2g^2\left(\tilde{\lambda}_{k,\chi}\right)^2\left(1+\tilde{m}_\phi^2\right)\xi_{\tilde{\phi}}}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right)\right]^3} \tag{2.73}$$

Here, one should emphasize those approximations become exact when $g = 0$. When one sets $g = 0$ in Eqs. (2.70 – 2.73) above, one returns exactly to the Defenu Picture where couplings are assumed inexistent; so algebraically the equations match the expectation that they should be equal to the decoupled picture when the coupling is removed. This gives a positive indication that the algebra carried-out and the derived equations are in fact correct. Furthermore, this simplification with taylor expansions at $g \to 0$ give another interpretation of the flow equations compared to the convoluted version previously described by Eqs. (2.63 – 2.66). Here, the coupled sector appears linearly, through interactions such as $g^2\tilde{\lambda}_{k,\phi}$ and $\left(g\tilde{\lambda}_{k,\phi}\right)^2$. Furthermore, outside of the cross-over $s^\star$ either $\tilde{m}_\chi^2$ or $\tilde{m}_\phi^2$ will become irrelevant parameters quickly; such that the equations linearize.

2.6 Momenta Dependencies and Anomalous Dimension

The anomalous dimension is evaluated analogous to the case of (for example) a scalar ϕ^4 theory, only that the propagator is in this case a matrix. By re-writing the propagator, Eq. (2.4), as Eq. (2.74),

$$P_k = \overline{P}_k \begin{pmatrix} \Gamma_{k,\chi}^{(2)} & -g_k \\ -g_k & \Gamma_{k,\phi}^{(2)} \end{pmatrix} = \frac{1}{\Gamma_{k,\phi}^{(2)}\Gamma_{k,\chi}^{(2)} - g_k^2} \begin{pmatrix} \Gamma_{k,\chi}^{(2)} & -g_k \\ -g_k & \Gamma_{k,\phi}^{(2)} \end{pmatrix} \tag{2.74}$$

One then calculates functional derivatives against the propagator, Eqs. (2.75, 2.76).

$$\frac{\delta P_k\left(p,q\right)}{\delta\phi\left(p_1\right)} = \int P_k\left(p,\xi\right) \begin{pmatrix} \Gamma_{k,\phi}^{(3)}\left(\xi,\zeta,p_1\right) & 0 \\ 0 & 0 \end{pmatrix} P_k\left(\zeta,q\right) d\xi d\zeta \tag{2.75}$$

$$
\begin{aligned}
\frac{\delta P_k(p,q)}{\delta\phi(p_1)\,\delta\phi(p_2)} &= 2\int P_k(p,\xi')\begin{pmatrix}\Gamma^{(3)}_{k,\phi}(\xi,\zeta,p_2) & 0\\ 0 & 0\end{pmatrix}P_k(\zeta',\xi)\begin{pmatrix}\Gamma^{(3)}_{k,\phi}(\xi,\zeta,p_1) & 0\\ 0 & 0\end{pmatrix}P_k(\zeta,q)\,d\xi d\zeta d\xi' d\zeta' \\
&= 2\int \overline{P_k}(p,\xi')\,\overline{P_k}(\zeta',\xi)\,\overline{P_k}(\zeta,q)\,\Gamma^{(3)}_{k,\phi}(\xi,\zeta,p_1)\,\Gamma^{(3)}_{k,\chi}(\xi',\zeta',p_2)\,d\xi d\zeta d\xi' d\zeta'\begin{pmatrix}\left(\Gamma^{(2)}_{k,\chi}\right)^3 & -g_k\left(\Gamma^{(2)}_{k,\chi}\right)^2\\ -g_k\left(\Gamma^{(2)}_{k,\chi}\right)^2 & g_k^2\Gamma^{(2)}_{k,\chi}\end{pmatrix} \\
&= 2\lambda_\phi^2\int\left(\overline{P_k}(p)\right)^2\overline{P_k}(p_2+r'-p)\,\phi(r)\,\phi(r')\,\delta(-2p+p_1+r)\,\delta(-2p+p_2+r')\,drdr'\,(\ldots) \\
&= 2\left(\lambda_\phi\phi_*\right)^2\left(\overline{P_k}(p)\right)^2\overline{P_k}(p_1-p)\begin{pmatrix}\left(\Gamma^{(2)}_{k,\chi}\right)^3 & -g_k\left(\Gamma^{(2)}_{k,\chi}\right)^2\\ -g_k\left(\Gamma^{(2)}_{k,\chi}\right)^2 & g_k^2\Gamma^{(2)}_{k,\chi}\end{pmatrix}
\end{aligned}
\tag{2.76}
$$

One could also conceive a non-zero anomalous dimension for the long-range node, Eq. (2.77).

$$
\begin{aligned}
\frac{\delta P_k(p,q)}{\delta\chi(p_1)\,\delta\chi(p_2)} &= 2\int P_k(p,\xi')\begin{pmatrix}0 & 0\\ 0 & \Gamma^{(3)}_{k,\chi}(\xi',\zeta',p_2)\end{pmatrix}P_k(\zeta',\xi)\begin{pmatrix}0 & 0\\ 0 & \Gamma^{(3)}_{k,\chi}(\xi',\zeta',p_2)\end{pmatrix}P_k(\zeta,q)\,d\xi d\zeta d\xi' d\zeta' \\
&= 2\int \overline{P_k}(p,\xi')\,\overline{P_k}(\zeta',\xi)\,\overline{P_k}(\zeta,q)\,\Gamma^{(3)}_{k,\phi}(\xi,\zeta,p_1)\,\Gamma^{(3)}_{k,\chi}(\xi',\zeta',p_2)\,d\xi d\zeta d\xi' d\zeta'\begin{pmatrix}g_k^2\Gamma^{(2)}_{k,\phi} & -g_k\left(\Gamma^{(2)}_{k,\phi}\right)^2\\ -g_k\left(\Gamma^{(2)}_{k,\phi}\right)^2 & \left(\Gamma^{(2)}_{k,\phi}\right)^3\end{pmatrix} \\
&= 2\lambda_\chi^2\int\left(\overline{P_k}(p)\right)^2\overline{P_k}(p_2+r'-p)\,\chi(r)\,\chi(r')\,\delta(-2p+p_1+r)\,\delta(-2p+p_2+r')\,drdr'\,(\ldots) \\
&= 2\left(\lambda_\chi\chi_*\right)^2\left(\overline{P_k}(p)\right)^2\overline{P_k}(p_1-p)\begin{pmatrix}g_k^2\Gamma^{(2)}_{k,\phi} & -g_k^2\left(\Gamma^{(2)}_{k,\phi}\right)^2\\ -g_k^2\left(\Gamma^{(2)}_{k,\phi}\right)^2 & \left(\Gamma^{(2)}_{k,\phi}\right)^3\end{pmatrix}
\end{aligned}
\tag{2.77}
$$

For the sake of completeness, in principle mixing could also incur momenta dependencies, Eq. (2.78).

$$
\begin{aligned}
\frac{\delta P_k(p,q)}{\delta\phi(p_1)\,\delta\chi(p_2)} &= 2\int P_k(p,\xi')\begin{pmatrix}0 & 0\\ 0 & \Gamma^{(3)}_{k,\chi}(\xi',\zeta',p_2)\end{pmatrix}P_k(\zeta',\xi)\begin{pmatrix}\Gamma^{(3)}_{k,\phi}(\xi,\zeta,p_1) & 0\\ 0 & 0\end{pmatrix}P_k(\zeta,q)\,d\xi d\zeta d\xi' d\zeta' \\
&= 2\int \overline{P_k}(p,\xi')\,\overline{P_k}(\zeta',\xi)\,\overline{P_k}(\zeta,q)\,\Gamma^{(3)}_{k,\phi}(\xi,\zeta,p_1)\,\Gamma^{(3)}_{k,\chi}(\xi',\zeta',p_2)\,d\xi d\zeta d\xi' d\zeta'\begin{pmatrix}g_k^2\Gamma^{(2)}_{k,\chi} & -g_k^3\\ -g_k\Gamma^{(2)}_{k,\phi}\Gamma^{(2)}_{k,\chi} & g_k^2\Gamma^{(2)}_{k,\phi}\end{pmatrix} \\
&= 2\lambda_\phi\lambda_\chi\int\left(\overline{P_k}(p)\right)^2\overline{P_k}(p_2+r'-p)\,\phi(r)\,\chi(r')\,\delta(-2p+p_1+r)\,\delta(-2p+p_2+r')\,drdr'\,(\ldots) \\
&= 2\lambda_\phi\lambda_\chi\phi_*\chi_*\left(\overline{P_k}(p)\right)^2\overline{P_k}(p_1-p)\begin{pmatrix}g_k^2\Gamma^{(2)}_{k,\chi} & -g_k^3\\ -g_k\Gamma^{(2)}_{k,\phi}\Gamma^{(2)}_{k,\chi} & g_k^2\Gamma^{(2)}_{k,\phi}\end{pmatrix}
\end{aligned}
\tag{2.78}
$$

In the picture suggested by Defenu, the Ansatz for the effective action, Eq. (2.79); where "..." indicates additional self- and mixing interactions; in principle allows for a non-zero anomalous dimension $Z_k^{\chi,2}$ for the long-ranged mode χ. In this projection, the evaluation of $\frac{\delta P_k(p,q)}{\delta\chi(p_1)\delta\chi(p_2)}$ in Eq. (2.77) above would lead to non-trivial anomalous dimension $Z_k^{\chi,2}$ for the long-ranged field.

$$\Gamma_k\left[\phi;\chi\right](p) = \frac{\chi^2}{2}\left(Z_k^{\chi,2}p^2 + Z_k^{\chi,s}p^s\right) + \frac{\phi^2}{2}Z_k^{\phi}p^2 + \dots \tag{2.79}$$

On the other hand, also consistent with Defenu picture, when $s < s^\star$, the p^s node dominates, and the RG flow drags the contribution that $Z_k^{\chi,2}$ is multiplied to to zero, because it is multiplied by powers of k tending to zero on the IR limit. In face of this justification one in principle ignores the contribution induced by Eq. (2.77); at least for an initial investigation of this problem.

In order to calculate the anomalous dimension, suffices to substitute Eq. (2.76) onto the Wetterich Equation, Eq. ().

$$\begin{aligned}
\frac{\delta^2\Gamma_k\left(p,q\right)}{\delta\phi\left(p_1\right)\delta\phi\left(p_2\right)} &= Tr\left(\left(\partial_k R_k\right)\frac{\delta P_k\left(p,q\right)}{\delta\phi\left(p_1\right)\delta\phi\left(p_2\right)}\right) = \\
&= 2\left(\lambda_\phi\phi_*\right)^2\left(\overline{P_k}\left(p\right)\right)^2\overline{P_k}\left(p_1-p\right)\left\{\left(\Gamma_{k,\chi}^{(2)}\right)^3\int\left(\partial_k R_{\phi,k}\right)dp + g_k^2\Gamma_{k,\chi}^{(2)}\int\left(\partial_k R_{\chi,k}\right)dp\right\} \\
&= 2\left(\lambda_\phi\phi_*\right)^2\left(\overline{P_k}\left(p\right)\right)^2\overline{P_k}\left(p_1-p\right)\left\{\left(\Gamma_{k,\chi}^{(2)}\right)^3 C_\phi Z_k^\phi + g_k^2\Gamma_{k,\chi}^{(2)}C_\chi Z_k^\chi k^s\right\}k^3
\end{aligned}$$

For $g \to 0$, one obtains the second contribution inside the brackets in Eq. (2.76) vanish, since it is proportional to g^2. On the other hand, one finds Eq. (2.80). This procedure algebraically matches the expectation that when $g = 0$ the anomalous dimension shoul dbe equal to the anomalous dimension in the decoupled picture. This gives a positive indication that the algebra carried-out and the derived equations are in fact correct.

$$\left(\overline{P_k}\left(p\right)\right)^2\overline{P_k}\left(p_1-p\right)\left(\Gamma_{k,\chi}^{(2)}\right)^3 = \left.\frac{\left(\Gamma_{k,\chi}^{(2)}\right)^3}{\left(\Gamma_{k,\phi}^{(2)}\Gamma_{k,\chi}^{(2)} - g_k^2\right)^2\Big|_p\left(\Gamma_{k,\phi}^{(2)}\Gamma_{k,\chi}^{(2)} - g_k^2\right)\Big|_{p-p_1}}\right|_{g=0} = \frac{1}{\left(\Gamma_{k,\phi}^{(2)}\right)^2\Big|_p\left(\Gamma_{k,\phi}^{(2)}\right)\Big|_{p-p_1}} \tag{2.80}$$

By now taylor-expanding $\overline{P_k}\left(p_1-p\right)$ around $p_1 \to 0$ and taking derivatives of $p_1$with respect to $\frac{\delta^2\Gamma_k(p,q)}{\delta\phi(p_1)\delta\phi(p_2)}$, one obtains Eq. (2.81) for the anomalous dimension.

$$\eta_\phi = \frac{2\left(\lambda_\phi \phi_*\right)^2}{\left(\Gamma^{(2)}_{k,\phi}\Gamma^{(2)}_{k,\chi} - g_k^2\right)^4}\left\{\left(\Gamma^{(2)}_{k,\chi}\right)^3 C_\phi Z_k^\phi + g_k^2 \Gamma^{(2)}_{k,\chi} C_\chi Z_k^\chi k^s\right\} k^3 \tag{2.81}$$

2.7 The Running of g_k

One calculates the flow of $\Gamma^{(2)}_{k,\phi\chi}$ with help of Eq. (2.78) in the Wetterich Equation, Eq. (2.82).

$$\begin{aligned}\frac{\delta^2 \Gamma_k\left(p,q\right)}{\delta\phi\left(p_1\right)\delta\chi\left(p_2\right)} &= Tr\left(\left(\partial_k R_k\right)\frac{\delta P_k\left(p,q\right)}{\delta\phi\left(p_1\right)\delta\chi\left(p_2\right)}\right) = \\ &= 2\lambda_\phi\lambda_\chi\phi_*\chi_* g_k^2\left(\overline{P_k}\left(p\right)\right)^3\left\{\Gamma^{(2)}_{k,\chi}\int\left(\partial_k R_{\phi,k}\right)dp + \Gamma^{(2)}_{k,\phi}\int\left(\partial_k R_{\chi,k}\right)dp\right\} \\ &= 2\lambda_\phi\lambda_\chi\phi_*\chi_* g_k^2\frac{\left[C_\chi\left(1+\tilde{m}_\phi^2\right) + C_\phi\left(1+\tilde{m}_\chi^2\right)\right]}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right) - g^2\right]^3}\end{aligned} \tag{2.82}$$

One furthermore introduces the contribution corresponding to making the left-handed side of the equation adimensionless, Eq. (2.83), which has been calculated previously in Eq. (2.58) already.

$$\mathcal{D}^{(1,1)}_{\phi\chi} = -\left(1+\frac{s-\eta_\chi-\eta_\phi}{2}\right)\tilde{U}^{(1,1)}_{t,\phi\chi} + \frac{\eta_\phi}{2}\tilde{\phi}\tilde{U}^{(2,1)}_{t,\phi\chi} + \frac{2-s+\eta_\chi}{2}\tilde{\chi}\tilde{U}^{(1,2)}_{t,\phi\chi} \tag{2.83}$$

The flow of $\tilde{g}_t$ is thus given by Eq. (2.84). One obtains a non-trivial flow for the coupling $\tilde{g}_t$, so long as the phase has non-zero wavefunctions ϕ_* and χ_*.

$$\partial_t\tilde{g}_t = -\left(1+\frac{s-\eta_\chi-\eta_\phi}{2}\right)\tilde{g}_t + 2\lambda_\phi\lambda_\chi\phi_*\chi_* g_t^2\frac{\left[C_\chi\left(1+\tilde{m}_\phi^2\right) + C_\phi\left(1+\tilde{m}_\chi^2\right)\right]}{\left[\left(1+\tilde{m}_\phi^2\right)\left(1+\tilde{m}_\chi^2\right) - g^2\right]^3} \tag{2.84}$$